

Fahad Bin Mohammad Hossain

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PROFESSIONAL SUMMARY

Backend Software Engineer promoted from Associate SE I to SE II within 12 months. Builds and operates scalable multi-tenant NestJS/PostgreSQL/MongoDB systems focused on event-driven architecture, secure authentication, rate limiting, and AI/LLM integrations (embeddings, RAG, frontier lab APIs) for production backend infrastructure at a global SaaS company.

EXPERIENCE

Associate Software Engineer II, 6sense HQ - Dhaka, Bangladesh Sep 2025 - Present

- Reduced business logic execution errors by 13% by building scalable backend APIs with retry mechanisms and async workflows across interconnected modules, using NestJS, PostgreSQL, MongoDB, and modular architecture.
- Increased data pipeline throughput significantly through event-driven processing pipelines that combine Voyage AI embeddings, Claude APIs, background workers, and queue-based messaging for higher fault tolerance.
- Strengthened multi-tenant security by implementing JWT/OAuth authentication with RBAC and distinct rate limits per tenant and per role, improving authorization performance under concurrent load.
- Cut deployment time by 20% by managing containerized CI/CD workflows with multi-stage Docker and AWS that support automated rollbacks across staging and production environments.

Associate Software Engineer I, 6sense HQ - Dhaka, Bangladesh Sep 2024 - Aug 2025

- Delivered 10% faster feature rollout by developing modular RESTful services in NestJS and PostgreSQL while enforcing clean coding standards and high unit test coverage across shared modules.
- Reduced query latency by 17% on high-traffic transactional workflows by optimizing PostgreSQL and MongoDB with targeted indexing, query refactoring, and efficient aggregation pipelines.
- Cut large file upload times by 60% by introducing S3 pre-signed URLs that let the frontend upload files in chunks outside the request-response cycle, improving reliability for large payloads.
- Improved deployment consistency by 20% by creating reusable validation, authentication, and testing components paired with modular service design and containerized CI/CD pipelines.

SKILLS

Languages	C++, JavaScript, TypeScript, Python
Databases	Postgres, MongoDB, Redis, VectorDB
Frameworks & Tools	Express, NestJS, NextJS, Docker, RabbitMQ, AWS (EC2, ECS, S3 etc)
AI Engineering	Retrieval Augmented Generation (RAG), LLM API Integration, Prompt Engineering
Architecture	Monolith, Microservices, Event Driven Design

PROFESSIONAL PROJECTS

- Pattern50 - Your startup studio for the AI era** - NestJS, NextJS, TypeScript, MongoDB, Claude, RabbitMQ, Redis, AWS (S3/ECR/EKS), K8s, 1Password, Google Drive Integration, Zoho Books, Stripe, Codacy, JWT/OAuth, Jest, Docker
 - Developed multi-tenant JWT/RBAC auth with access levels (Global / Studio / Company), invite onboarding, refresh tokens, and role-permission scoping for studio and company portfolio management across the platform.
 - Connected Google Drive OAuth and GitHub App integrations for document and code ingestion into AI knowledge bases, with Bull-based token refresh, async embedding jobs, and credentials stored in 1Password vaults.
 - Designed a RAG pipeline with Voyage embeddings, document chunking, and MongoDB Atlas vector search, plus an OpenAI-powered report generator to turn company knowledge into scoped AI answers and reports.
 - Built a Claude AI agent with multi-round tool-calling (GitHub + Codacy) and SSE streaming chat so studios can query repositories, pull requests, workflows, and code-quality signals in real time.
 - Built Stripe billing (PaymentIntents, cards, ACH, webhooks, Socket.IO events) and Zoho Books invoicing over event driven NestJS microservices (RabbitMQ, Redis, Bull), deployed via Docker to AWS ECR/EKS with GitOps CI/CD.
- Jeter AI - AI powered dating, compatibility & matchmaking platform** - NestJS, NextJS, TypeScript, PostgreSQL, Prisma ORM, OpenAI, Redis (Valkey), AWS (S3/SES/ECS/ECR), Agora, Stripe, JWT/OAuth, Jest, Docker
 - Architected a scalable, modular NestJS backend with 15+ domain modules using clean layered architecture, dependency injection, and reusable service patterns, enabling maintainable code organization and faster feature delivery across teams.
 - Developed an AI-driven compatibility engine using OpenAI APIs and token-based embeddings to generate personalized heart, mind, and body scores along with dynamic Jeter summaries from structured user assessment responses.
 - Implemented secure multi-provider authentication supporting JWT, Google OAuth, and OTP login with authorization guards, CSRF protection, rate limiting, and security headers to ensure safe and compliant user access workflows.
 - Integrated presigned AWS S3 uploads, Agora messaging/video, Stripe payments, and Redis caching, while maintaining reliability through Jest unit testing, Swagger API documentation, and CI/CD pipelines with Jenkins and GitHub Actions.

EDUCATION

Bangladesh University of Business and Technology 2019 – 2023
Bachelor of Computer Science and Engineering — CGPA: 3.63
Relevant Coursework: OS, DSA, AI, ML, Computer Networks, Database, Discrete Mathematics, Microprocessor & Microcontroller etc.